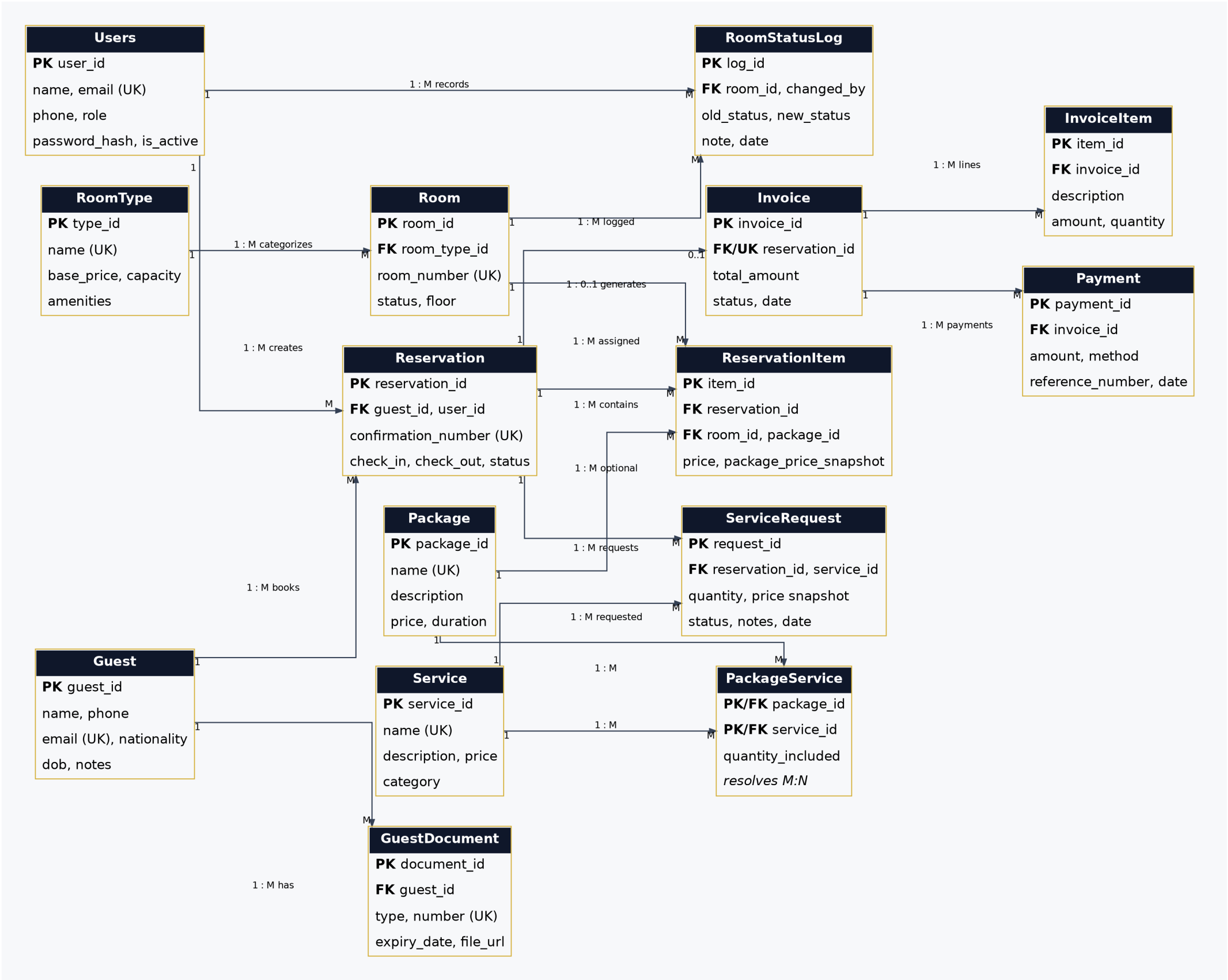


Hotel Management System - ER Diagram

Actual schema: 15 entities with keys, constraints, and cardinalities



Relationship Legend and Cardinality Map

Supporting notes for the ER diagram and the implemented MySQL schema

Symbol / Label	Meaning
PK	Primary key
FK	Foreign key
UK	Unique key
1 : M	One parent record can relate to many child records
1 : 0..1	One parent record can have zero or one child record
M:N resolved	Many-to-many relationship implemented through PackageService

Core Relationships

Relationship	Cardinality	Purpose
Users -> Reservation	1:M	A staff user can create many reservations.
Users -> RoomStatusLog	1:M	A staff user can record many room status changes.
Guest -> Reservation	1:M	One guest may have many bookings.
Guest -> GuestDocument	1:M	One guest may have multiple identity documents.
RoomType -> Room	1:M	One room type can categorize many rooms.
Room -> ReservationItem	1:M	A room can appear in many reservation items over time.
Room -> RoomStatusLog	1:M	One room can have many status log records.
Reservation -> ReservationItem	1:M	A reservation can contain one or more room/package lines.
Reservation -> ServiceRequest	1:M	A reservation can have many service requests.
Reservation -> Invoice	1:0..1	An invoice is generated at checkout, so a reservation may have no invoice before checkout.
Package -> ReservationItem	1:M	A package may be selected in many reservation items.
Package + Service -> PackageService	M:N resolved	PackageService resolves the many-to-many relationship between packages and services.
Service -> ServiceRequest	1:M	A catalog service can be requested many times.
Invoice -> InvoiceItem	1:M	One invoice contains multiple billing line items.
Invoice -> Payment	1:M	One invoice can receive partial or full payments.